

Qualification Pack



Green Hydrogen Plant Maintenance Technician

QP Code: HYC/Q4007

Version: 1.0

NSQF Level: 4

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HYC/Q4007: Green Hydrogen Plant Maintenance Technician

Brief Job Description

The green hydrogen maintenance technician plays a crucial part in the seamless operation and maintenance of the hydrogen production, storage, and distribution systems. Responsibilities encompass a comprehensive spectrum of maintenance activities, from routine inspections and troubleshooting to predictive strategies utilizing advanced data analysis tools. Ensuring safety protocols are strictly adhered to, the role involves proactive responses to emergencies and participation in safety drills, meticulous record-keeping, optimizing efficiency, reducing costs, and enhancing overall safety within the hydrogen production facility.

Personal Attributes

The individual should have ability to give attention to detail, technical aptitude, adaptability, teamwork and a strong commitment to safety. The person should have the ability to work individually / team round the clock and must always promote safety. Must be able to stay calm in a fast-paced environment where they may be expected to meet production targets of the hydrogen generation units. This is a hands-on position that requires physical dexterity and attention to detail.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4001: Hydrogen Safety, Security and Health Management Procedures](#)
2. [HYC/N4014: Introduction to Green hydrogen and value chain](#)
3. [HYC/N4015: Water Electrolyser systems](#)
4. [HYC/N4016: Green Hydrogen Storage systems](#)
5. [HYC/N4017: HSE and Commissioning Checklist](#)
6. [HYC/N4018: Roles & Responsibilities](#)
7. [HYC/N9301: Working effectively in a team](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
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Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
Country	India
NSQF Level	4
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO/2015-3122.1400
Minimum Educational Qualification & Experience	Completed 2nd year of the 3-year diploma after 10 with 1 Year of experience OR 10th grade pass with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	29/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-04-HY-02523-2024-V1-HSSCI
NQR Version	1.0

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HYC/N4001: Hydrogen Safety, Security and Health Management Procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace where Hydrogen is produced and handled.

Scope

The scope covers the following :

- Understanding Properties of Hydrogen from safety point of view
- Safe handling of Hydrogen
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.
- PC2.** Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions
- PC3.** Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.
- PC4.** Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.
- PC5.** Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire
- PC6.** capability to accurately read and interpret data from sensing equipment used in hydrogen production processes
- PC7.** apply key safety parameters for handling hydrogen & oxygen
- PC8.** Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames
- PC9.** Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards
- PC10.** competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace
- PC11.** Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations
- PC12.** Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.

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- PC13.** safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries
- PC14.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC15.** Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.
- PC16.** Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.
- PC17.** Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.

Ensure the following fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC18.** Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard
- PC20.** Promote and enforce following good housekeeping practice in order to prevent fire hazards
- PC21.** Regularly inspect work areas for compliance with housekeeping standards
- PC22.** identify and document issues concerning the hydrogen safety in work place
- PC23.** Collaborate with relevant departments to address and rectify safety concerns promptly
- PC24.** Ensure that work areas and escape routes are well maintained and free from any hindrance
- PC25.** Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation
- PC26.** Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard
- PC27.** Ensure that personnel are trained in preparing comprehensive accident/incident reports
- PC28.** Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC29.** provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues
- PC30.** Be proficient in basic techniques of bandaging
- PC31.** Lead by example in responding promptly and appropriately to an accident
- PC32.** perform rescue activity during an accident in real or simulated environments
- PC33.** Familiarize personnel with correct escape routes during an emergency
- PC34.** demonstrate correct method to rescue injured people and others during an emergency
- PC35.** Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen</i>	17	27	-	-
PC1. Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.	1	2	-	-
PC2. Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions	1	1	-	-
PC3. Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.	1	2	-	-
PC4. Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.	1	2	-	-
PC5. Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire	1	2	-	-
PC6. capability to accurately read and interpret data from sensing equipment used in hydrogen production processes	1	2	-	-
PC7. apply key safety parameters for handling hydrogen & oxygen	1	2	-	-
PC8. Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames	1	1	-	-
PC9. Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards	1	1	-	-
PC10. competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations	1	1	-	-
PC12. Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.	1	1	-	-
PC13. safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries	1	2	-	-
PC14. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC15. Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.	1	2	-	-
PC16. Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.	1	1	-	-
PC17. Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.	1	1	-	-
<i>Ensure the following fire safety procedures</i>	11	19	-	-
PC18. Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly	1	2	-	-
PC19. Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard	1	2	-	-
PC20. Promote and enforce following good housekeeping practice in order to prevent fire hazards	1	2	-	-
PC21. Regularly inspect work areas for compliance with housekeeping standards	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify and document issues concerning the hydrogen safety in work place	1	2	-	-
PC23. Collaborate with relevant departments to address and rectify safety concerns promptly	1	2	-	-
PC24. Ensure that work areas and escape routes are well maintained and free from any hindrance	1	2	-	-
PC25. Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation	1	2	-	-
PC26. Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard	1	1	-	-
PC27. Ensure that personnel are trained in preparing comprehensive accident/incident reports	1	1	-	-
PC28. Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions	1	1	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	7	14	-	-
PC29. provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues	1	2	-	-
PC30. Be proficient in basic techniques of bandaging	1	2	-	-
PC31. Lead by example in responding promptly and appropriately to an accident	1	2	-	-
PC32. perform rescue activity during an accident in real or simulated environments	1	2	-	-
PC33. Familiarize personnel with correct escape routes during an emergency	1	2	-	-
PC34. demonstrate correct method to rescue injured people and others during an emergency	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.	1	2	-	-
NOS Total	35	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4001
NOS Name	Hydrogen Safety, Security and Health Management Procedures
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4014: Introduction to Green hydrogen and value chain

Description

This unit covers overview of hydrogen molecule as an energy carrier and its basic properties, along with complete value chain of Green Hydrogen including Production, Storage, Transportation, Applications.

Scope

The scope covers the following :

- Basics of hydrogen and Green Hydrogen
- Typical Green Hydrogen Plant and its layout
- Power sources for Green Hydrogen Plant

Elements and Performance Criteria

Understanding Hydrogen molecule as an energy carrier

To be competent, the user/individual on the job must be able to:

- PC1.** Understand necessity of green hydrogen
- PC2.** Discuss properties and characteristics of Hydrogen
- PC3.** Discuss key aspects related to Green Hydrogen production, transportation, storage and applications
- PC4.** Discuss the role and responsibilities of a Green Hydrogen Plant Technician and importance of doing this course
- PC5.** Explain the necessity of green hydrogen in sustainable energy transition
- PC6.** Discuss the properties and characteristics of Hydrogen
- PC7.** Describe basic concepts of Hydrogen as an energy carrier
- PC8.** Discuss various colour code nomenclature of Hydrogen
- PC9.** Discuss the value chain for Green Hydrogen
- PC10.** Discuss benefits and drawbacks of existing methods of Hydrogen production
- PC11.** Discuss key aspects and challenges related to production, storage, transportation, distribution of Green Hydrogen
- PC12.** Discuss the end use applications of Green hydrogen in industry, transport and power production
- PC13.** Discuss briefly various technology options for production of green hydrogen
- PC14.** Discuss various renewable energy sources for large scale production of green hydrogen through electrolysis of water
- PC15.** Discuss Green hydrogen economy in Indian context
- PC16.** Discuss the key aspects of Green Hydrogen Policy and Mission of Government of India
- PC17.** Discuss the role and responsibilities of a Green Hydrogen Plant Technician
- PC18.** Identify key components of the Green Hydrogen plant including electrical, mechanical and civil components

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- PC19.** Discuss functions of the key components including of electrolyzer stacks, renewable power supply system, feed water supply and demineralisation system, gas separator, transformer and rectifier, gas compression unit, etc.
- PC20.** Discuss fundamental principles of main components on which they operate (e.g. electrolyzer stack, gas collector, power source, etc.)
- PC21.** Types of electrolyzers: brief description
- PC22.** Explain overall layout of the plant
- PC23.** Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area
- PC24.** Explain how to read and interpret electrical and other applicable codes, standards and protocols relevant to the plant
- PC25.** Explain key material and safety codes, Technology protocols and standards as applicable in Green hydrogen industry
- PC26.** Discuss the key parameters of renewable and hybrid power plants
- PC27.** Explain how cost effective and reliable Renewable Power can be assured for the production of Green Hydrogen
- PC28.** Discuss the flexible system operation for the supply of power from different sources
- PC29.** Discuss how Sizing of renewable power plant along with storage capacity is critical to meet hydrogen load demand
- PC30.** Discuss functions of transformers and rectifiers
- PC31.** Explain key aspects of Maintaining stability of power supply for the green hydrogen plant, switching of number of electrolyser cells as per availability of electric power

Green hydrogen production pathways and purification

To be competent, the user/individual on the job must be able to:

- PC32.** Know hydrogen production using conventional routes
- PC33.** Understand green hydrogen production using renewable energy sources like solar & wind
- PC34.** Identify and know different types of electrolyzers
- PC35.** Understand green hydrogen production from biomass gasification
- PC36.** Know purification requirements of hydrogen
- PC37.** Comprehend emissions from different hydrogen production methods
- PC38.** Understand basis for different colours of hydrogen

Hydrogen compression, storage and transportation

To be competent, the user/individual on the job must be able to:

- PC39.** Know the need for hydrogen compression
- PC40.** Understand physical storage of hydrogen in cylinders and vessels
- PC41.** Know why different types of storage cylinders are required
- PC42.** Understand chemical storage of hydrogen
- PC43.** Know how hydrogen is transported in different ways

Applications of Hydrogen

To be competent, the user/individual on the job must be able to:

- PC44.** Know hydrogen applications and quality requirements.
- PC45.** Understand application of hydrogen as a blend with CNG

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PC46. know application of hydrogen in Fuel Cell

PC47. Observe application of hydrogen in direct IC engine

PC48. List down application of hydrogen in different industries

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the properties and use of green hydrogen

KU2. Differences between hydrogen and other fuels as energy carrier

KU3. basis for different colours of hydrogen

KU4. Observe significance of green hydrogen value chain

KU5. Emissions from each of the production methods

KU6. principles and processes involved in hydrogen production

KU7. Different ways of hydrogen storage- gaseous, liquid, solid-state storage

KU8. pros and cons of each storage method

KU9. Ways of transportation of hydrogen

KU10. Safety protocols in handling Hydrogen

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively and share the information efficiently

GS2. Effective use of computers (MS office tools)

GS3. Presentation skills -oral and written.

GS4. Effective use read and interpret the data

GS5. read instructions, guidelines/procedures and reports

GS6. analyze the parameters and output

GS7. communicate effectively with team members.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Hydrogen molecule as an energy carrier</i>	62	31	-	-
PC1. Understand necessity of green hydrogen	2	1	-	-
PC2. Discuss properties and characteristics of Hydrogen	2	1	-	-
PC3. Discuss key aspects related to Green Hydrogen production, transportation, storage and applications	2	1	-	-
PC4. Discuss the role and responsibilities of a Green Hydrogen Plant Technician and importance of doing this course	2	1	-	-
PC5. Explain the necessity of green hydrogen in sustainable energy transition	2	1	-	-
PC6. Discuss the properties and characteristics of Hydrogen	2	1	-	-
PC7. Describe basic concepts of Hydrogen as an energy carrier	2	1	-	-
PC8. Discuss various colour code nomenclature of Hydrogen	2	1	-	-
PC9. Discuss the value chain for Green Hydrogen	2	1	-	-
PC10. Discuss benefits and drawbacks of existing methods of Hydrogen production	2	1	-	-
PC11. Discuss key aspects and challenges related to production, storage, transportation, distribution of Green Hydrogen	2	1	-	-
PC12. Discuss the end use applications of Green hydrogen in industry, transport and power production	2	1	-	-
PC13. Discuss briefly various technology options for production of green hydrogen	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Discuss various renewable energy sources for large scale production of green hydrogen through electrolysis of water	2	1	-	-
PC15. Discuss Green hydrogen economy in Indian context	2	1	-	-
PC16. Discuss the key aspects of Green Hydrogen Policy and Mission of Government of India	2	1	-	-
PC17. Discuss the role and responsibilities of a Green Hydrogen Plant Technician	2	1	-	-
PC18. Identify key components of the Green Hydrogen plant including electrical, mechanical and civil components	2	1	-	-
PC19. Discuss functions of the key components including of electrolyzer stacks, renewable power supply system, feed water supply and demineralisation system, gas separator, transformer and rectifier, gas compression unit, etc.	2	1	-	-
PC20. Discuss fundamental principles of main components on which they operate (e.g. electrolyzer stack, gas collector, power source, etc.)	2	1	-	-
PC21. Types of electrolyzers: brief description	2	1	-	-
PC22. Explain overall layout of the plant	2	1	-	-
PC23. Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area	2	1	-	-
PC24. Explain how to read and interpret electrical and other applicable codes, standards and protocols relevant to the plant	2	1	-	-
PC25. Explain key material and safety codes, Technology protocols and standards as applicable in Green hydrogen industry	2	1	-	-
PC26. Discuss the key parameters of renewable and hybrid power plants	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. Explain how cost effective and reliable Renewable Power can be assured for the production of Green Hydrogen	2	1	-	-
PC28. Discuss the flexible system operation for the supply of power from different sources	2	1	-	-
PC29. Discuss how Sizing of renewable power plant along with storage capacity is critical to meet hydrogen load demand	2	1	-	-
PC30. Discuss functions of transformers and rectifiers	2	1	-	-
PC31. Explain key aspects of Maintaining stability of power supply for the green hydrogen plant, switching of number of electrolyser cells as per availability of electric power	2	1	-	-
<i>Green hydrogen production pathways and purification</i>	14	5	-	-
PC32. Know hydrogen production using conventional routes	2	-	-	-
PC33. Understand green hydrogen production using renewable energy sources like solar & wind	2	1	-	-
PC34. Identify and know different types of electrolyzers	2	1	-	-
PC35. Understand green hydrogen production from biomass gasification	2	1	-	-
PC36. Know purification requirements of hydrogen	2	-	-	-
PC37. Comprehend emisisions from different hydrogen production methods	2	1	-	-
PC38. Understand basis for different colours of hydrogen	2	1	-	-
<i>Hydrogen compression, storage and transportation</i>	10	4	-	-
PC39. Know the need for hydrogen compression	2	1	-	-
PC40. Understand physical storage of hydrogen in cylinders and vessels	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC41. Know why different types of storage cylinders are required	2	-	-	-
PC42. Understand chemical storage of hydrogen	2	1	-	-
PC43. Know how hydrogen is transported in different ways	2	1	-	-
<i>Applications of Hydrogen</i>	9	5	-	-
PC44. Know hydrogen applications and quality requirements.	1	1	-	-
PC45. Understand application of hydrogen as a blend with CNG	2	1	-	-
PC46. know application of hydrogen in Fuel Cell	2	1	-	-
PC47. Observe application of hydrogen in direct IC engine	2	1	-	-
PC48. List down application of hydrogen in different industries	2	1	-	-
NOS Total	95	45	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4014
NOS Name	Introduction to Green hydrogen and value chain
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4015: Water Electrolyser systems

Description

This unit covers the production of green hydrogen using electrolyser technology, different types of electrolysers, importance of water purification unit and hydrogen purification and compression systems.

Scope

The scope covers the following :

- Electrolyser technology for Green Hydrogen Production
- Water feed and purification unit
- Hydrogen purification and compression

Elements and Performance Criteria

Electrolyser technology for Green Hydrogen Production

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss the types of Electrolysers including Polymer Electrolyte Membrane (PEM), Alkaline electrolysis (AE), Solid Oxide Electrolysis (SOEC), Anion Exchange Membranes (AEM)
- PC2.** Explain the key technical specifications of various Electrolyser types (PEM, AE AEM, SOEC) including capacity and sizing
- PC3.** Provide an overview of various electrolyser along with their key features and comparison
- PC4.** Explain the major components of an Electrolyser
- PC5.** Discuss the Operating principles of PEM AE and SOEC electrolyser.
- PC6.** Identify suitable tools and equipment required for Installation of electrolyser
- PC7.** Understand relevant technical sheets, safety and technical standards for proper execution of work.
- PC8.** Explain the inputs / outputs of an Electrolyser system
- PC9.** Explain key O&M requirements for electrolyzer
- PC10.** Know how to calculate losses and equipment efficiency
- PC11.** Discuss basic Technical specifications of electrolyser
- PC12.** Read and interpret drawings, mechanical and electrical diagrams, Plant specifications
- PC13.** Explain step by step process of installation of electrolyzer
- PC14.** Read and interpret drawings, Mechanical and electrical diagrams, Plant specifications
- PC15.** Know how to install the plant using design drawings and documents provided
- PC16.** Discuss in detail the input renewable power from various sources and its integration with electrolyzer.
- PC17.** Identify assembly/Installation of Parts and Components of Electrolyzer
- PC18.** Understand how to perform testing, operation, maintaining and troubleshooting of an electrolyzer
- PC19.** Discuss Key selection parameters of electrolysis including density, pressure, operating temperature, hydrogen purity

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PC20. Know how to Maintain the log for daily operation of electrolyzer

PC21. Explain Start up, shutdown, and operate the production processes

Installation of water feed system

To be competent, the user/individual on the job must be able to:

PC22. Explain specifications of water required for electrolyser

PC23. Know the schematic of input water system and treatment required

PC24. Discuss various parameters for feed water supply

PC25. Record water consumption in electrolyser and loss/wastage of water

PC26. Understand the various processes for treating the water supply for feedstock purpose

PC27. Identify various parameters essential for water as feedstock

PC28. Know selection of water quality for hydrogen production

PC29. monitor the quality of water feedstock for input to the electrolyzer.

PC30. Describe layout of piping system for supply of water to electrolyzer

PC31. install the Piping and Piping accessories for supplying controlled quantity of water to the electrolyzer using level controller etc.

PC32. Understand how to record results, organize data, and perform basic computations to set up and operate input water system

PC33. List down do's and don'ts in installation of water supply unit and its piping and joints

Hydrogen purification and Compression

To be competent, the user/individual on the job must be able to:

PC34. Explain the process of conditioning/purification of green hydrogen

PC35. Describe the working n De-Oxo System for removal of oxygen from hydrogen.

PC36. Know working of Dryer System.

PC37. set up and operate the hydrogen conditioning system

PC38. Identify other key balance of plants components

PC39. Explain compression process

PC40. install compression system

PC41. Identify H₂ compression types such as reciprocating, ionic & diaphragm.

PC42. read and interpret Material Safety Data Sheet (MSDS)

PC43. select and install hydrogen compression system

PC44. Discuss the precautions required to compress hydrogen

PC45. Explain key components/ parameters of compressors including Pressure Valve, indicators, Gas Outlet & Inlets Valves, Tank size and Placement etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the properties and use of green hydrogen

KU2. Differences between hydrogen and other fuels as energy carrier

KU3. basis for different colours of hydrogen

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- KU4.** Observe significance of green hydrogen value chain
- KU5.** Emissions from each of the production methods
- KU6.** principles and processes involved in hydrogen production
- KU7.** Different ways of hydrogen storage- gaseous, liquid, solid-state storage
- KU8.** pros and cons of each storage method
- KU9.** Ways of transportation of hydrogen
- KU10.** Safety protocols in handling Hydrogen

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively and share the information efficiently
- GS2.** Effective use of computers (MS office tools)
- GS3.** Presentation skills -oral and written.
- GS4.** Effective use read and interpret the data
- GS5.** read instructions, guidelines/procedures and reports
- GS6.** analyze the parameters and output
- GS7.** communicate effectively with team members.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electrolyser technology for Green Hydrogen Production</i>	42	21	-	-
PC1. Discuss the types of Electrolysers including Polymer Electrolyte Membrane (PEM), Alkaline electrolysis (AE), Solid Oxide Electrolysis (SOEC), Anion Exchange Membranes (AEM)	2	1	-	-
PC2. Explain the key technical specifications of various Electrolyser types (PEM, AE AEM, SOEC) including capacity and sizing	2	1	-	-
PC3. Provide an overview of various electrolyser along with their key features and comparison	2	1	-	-
PC4. Explain the major components of an Electrolyser	2	1	-	-
PC5. Discuss the Operating principles of PEM AE and SOEC electrolyser.	2	1	-	-
PC6. Identify suitable tools and equipment required for Installation of electrolyser	2	1	-	-
PC7. Understand relevant technical sheets, safety and technical standards for proper execution of work.	2	1	-	-
PC8. Explain the inputs / outputs of an Electrolyser system	2	1	-	-
PC9. Explain key O&M requirements for electrolyzer	2	1	-	-
PC10. Know how to calculate losses and equipment efficiency	2	1	-	-
PC11. Discuss basic Technical specifications of electrolyser	2	1	-	-
PC12. Read and interpret drawings, mechanical and electrical diagrams, Plant specifications	2	1	-	-
PC13. Explain step by step process of installation of electrolyzer	2	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Read and interpret drawings, Mechanical and electrical diagrams, Plant specifications	2	1	-	-
PC15. Know how to install the plant using design drawings and documents provided	2	1	-	-
PC16. Discuss in detail the input renewable power from various sources and its integration with electrolyzer.	2	1	-	-
PC17. Identify assembly/Installation of Parts and Components of Electrolyzer	2	1	-	-
PC18. Understand how to perform testing, operation, maintaining and troubleshooting of an electrolyzer	2	1	-	-
PC19. Discuss Key selection parameters of electrolysis including density, pressure, operating temperature, hydrogen purity	2	1	-	-
PC20. Know how to Maintain the log for daily operation of electrolyzer	2	1	-	-
PC21. Explain Start up, shutdown, and operate the production processes	2	1	-	-
<i>Installation of water feed system</i>	24	12	-	-
PC22. Explain specifications of water required for electrolyser	2	1	-	-
PC23. Know the schematic of input water system and treatment required	2	1	-	-
PC24. Discuss various parameters for feed water supply	2	1	-	-
PC25. Record water consumption in electrolyser and loss/wastage of water	2	1	-	-
PC26. Understand the various processes for treating the water supply for feedstock purpose	2	1	-	-
PC27. Identify various parameters essential for water as feedstock	2	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Know selection of water quality for hydrogen production	2	1	-	-
PC29. monitor the quality of water feedstock for input to the electrolyzer.	2	1	-	-
PC30. Describe layout of piping system for supply of water to electrolyzer	2	1	-	-
PC31. install the Piping and Piping accessories for supplying controlled quantity of water to the electrolyzer using level controller etc.	2	1	-	-
PC32. Understand how to record results, organize data, and perform basic computations to set up and operate input water system	2	1	-	-
PC33. List down do's and don'ts in installation of water supply unit and its piping and joints	2	1	-	-
<i>Hydrogen purification and Compression</i>	24	12	-	-
PC34. Explain the process of conditioning/purification of green hydrogen	2	1	-	-
PC35. Describe the working n De-Oxo System for removal of oxygen from hydrogen.	2	1	-	-
PC36. Know working of Dryer System.	2	1	-	-
PC37. set up and operate the hydrogen conditioning system	2	1	-	-
PC38. Identify other key balance of plants components	2	1	-	-
PC39. Explain compression process	2	1	-	-
PC40. install compression system	2	1	-	-
PC41. Identify H ₂ compression types such as reciprocating, ionic & diaphragm.	2	1	-	-
PC42. read and interpret Material Safety Data Sheet (MSDS)	2	1	-	-
PC43. select and install hydrogen compression system	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC44. Discuss the precautions required to compress hydrogen	2	1	-	-
PC45. Explain key components/ parameters of compressors including Pressure Valve, indicators, Gas Outlet & Inlets Valves, Tank size and Placement etc	2	1	-	-
NOS Total	90	45	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4015
NOS Name	Water Electrolyser systems
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

Qualification Pack

HYC/N4016: Green Hydrogen Storage systems

Description

This unit covers various types of hydrogen storage systems and associated details in terms of leakage detection and safety aspects.

Scope

The scope covers the following :

- Various Hydrogen storage systems
- Hydrogen filling and depressurization
- Leakage detection and Safety aspects

Elements and Performance Criteria

Green Hydrogen Storage systems

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the need for hydrogen storage
- PC2.** Explain the Challenges associated with Hydrogen in storage, handling and transportation
- PC3.** Discuss how to select and install hydrogen storage system
- PC4.** List down precautions required to store hydrogen
- PC5.** Follow safety guidelines to be followed as per applicable standard
- PC6.** Know the architecture of Piping arrangement
- PC7.** Know basics of Storage layout requirement
- PC8.** Follow different methods of H₂ storage (single vessel/multi cylinder cascade/portable storage/ tube-trailers etc
- PC9.** Know how these storage systems are set up.
- PC10.** Identify type I/II/III/IV cylinders and difference in setting up method.
- PC11.** Outline use of special tools and tackles.
- PC12.** Understand depressurization methods of Hydrogen Storage System - Use of Water seal or vent stack system.
- PC13.** Identify key safety measures to be followed at hydrogen compression and storage system
- PC14.** Perform periodic / regular leak testing of Hydrogen storage Systems
- PC15.** carry out Gas leak detection & method of placement
- PC16.** understand the use of Firefighting system,
- PC17.** use Flame Resistant Clothing in Hydrogen area.
- PC18.** Discuss various safety related issues- Do's & Don'ts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** procedures and instructions specific to the area of assignment
- KU2.** industrial maintenance practices, preferably in hydrogen production or a similar process industry
- KU3.** troubleshooting of machinery and equipment associated with hydrogen production, storage, and distribution.
- KU4.** implementation and optimization of predictive maintenance programs using advanced data analysis tools.
- KU5.** cryogenic component maintenance, including routine inspections and preventive measures.
- KU6.** safety protocols, emergency response, and hydrogen handling procedures.
- KU7.** equipment alignment and calibration.
- KU8.** collaboration with cross-functional teams
- KU9.** importance of effective communication with stakeholders.
- KU10.** record-keeping, report generation, and meticulous documentation updates.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use Hand and Power Tools
- GS2.** communicate effectively and share the information efficiently
- GS3.** Effective use of computers (MS office tools)
- GS4.** Presentation skills -oral and written.
- GS5.** Effective use read and interpret the data
- GS6.** read instructions, guidelines/procedures and reports
- GS7.** analyze the parameters and output
- GS8.** communicate effectively with team members.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Green Hydrogen Storage systems</i>	40	20	-	-
PC1. Identify the need for hydrogen storage	2	1	-	-
PC2. Explain the Challenges associated with Hydrogen in storage, handling and transportation	3	1	-	-
PC3. Discuss how to select and install hydrogen storage system	3	2	-	-
PC4. List down precautions required to store hydrogen	3	1	-	-
PC5. Follow safety guidelines to be followed as per applicable standard	2	1	-	-
PC6. Know the architecture of Piping arrangement	2	1	-	-
PC7. Know basics of Storage layout requirement	2	1	-	-
PC8. Follow different methods of H ₂ storage (single vessel/multi cylinder cascade/portable storage/ tube-trailers etc	2	1	-	-
PC9. Know how these storage systems are set up.	2	1	-	-
PC10. Identify type I/II/III/IV cylinders and difference in setting up method.	2	1	-	-
PC11. Outline use of special tools and tackles.	2	1	-	-
PC12. Understand depressurization methods of Hydrogen Storage System - Use of Water seal or vent stack system.	2	1	-	-
PC13. Identify key safety measures to be followed at hydrogen compression and storage system	2	1	-	-
PC14. Perform periodic / regular leak testing of Hydrogen storage Systems	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. carry out Gas leak detection & method of placement	2	2	-	-
PC16. understand the use of Firefighting system,	2	1	-	-
PC17. use Flame Resistant Clothing in Hydrogen area.	2	1	-	-
PC18. Discuss various safety related issues- Do's & Don'ts	3	1	-	-
NOS Total	40	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4016
NOS Name	Green Hydrogen Storage systems
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4
Credits	2.5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

Qualification Pack

HYC/N4017: HSE and Commissioning Checklist

Description

This unit covers the detailed checklist to be followed by a technician for the commissioning of a green hydrogen plant, along with roles & responsibilities of the technician.

Scope

The scope covers the following :

- Health, Safety & Environment
- Commissioning Checklist

Elements and Performance Criteria

Health, Safety & Environment

To be competent, the user/individual on the job must be able to:

- PC1.** Perform Health and safety measures for installing and operating Green hydrogen system
- PC2.** Identify the requirements for safe work area at hydrogen generation project site
- PC3.** Understand health & Safety standards, procedures and regulations applicable for safe operation of hydrogen systems
- PC4.** Identify potential sources & causes of gas leaks, fire, explosion, bomb threatening, natural calamities etc.
- PC5.** Implement different detectors and safety tools
- PC6.** Review the Material Safety Data Sheet
- PC7.** maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.
- PC8.** Utilize sensors to alert the safety officer and other users when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals
- PC9.** Know how to administered first aid.
- PC10.** Identify the personal protective equipment used for the specific purpose.
- PC11.** Identify the hazards associated with hydrogen generation system
- PC12.** Carry out mock testing of firefighting system
- PC13.** discuss all applicable statutory requirements along with safety regulations in terms of fire protection.
- PC14.** Incorporate good housekeeping practices and infection control guidelines.

Commissioning Checklist

To be competent, the user/individual on the job must be able to:

- PC15.** Appreciate the importance of Commissioning activities
- PC16.** show how required activities according to commissioning checklist are performed
- PC17.** Know dismantling and removal of components
- PC18.** use the checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendations

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- PC19.** explain Green hydrogen system installation, testing and commissioning checklist
- PC20.** Identify various Inputs, outputs and key performance metrics for hydrogen generation
- PC21.** Implement all pre-commissioning tests and other aspects including hydrogen purity, pressure test, leakage
- PC22.** Make compressor Assembly operational
- PC23.** Perform & test Drying Plant Hydraulic
- PC24.** Carry out safety tests
- PC25.** Perform a trial run before the commercial runs
- PC26.** commission Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards/SOPs etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** procedures and instructions specific to the area of assignment
- KU2.** industrial maintenance practices, preferably in hydrogen production or a similar process industry
- KU3.** troubleshooting of machinery and equipment associated with hydrogen production, storage, and distribution.
- KU4.** implementation and optimization of predictive maintenance programs using advanced data analysis tools.
- KU5.** cryogenic component maintenance, including routine inspections and preventive measures
- KU6.** safety protocols, emergency response, and hydrogen handling procedures.
- KU7.** equipment alignment and calibration.
- KU8.** collaboration with cross-functional teams
- KU9.** importance of effective communication with stakeholders.
- KU10.** record-keeping, report generation, and meticulous documentation updates.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use Hand and Power Tools
- GS2.** communicate effectively and share the information efficiently
- GS3.** Effective use of computers (MS office tools)
- GS4.** Presentation skills -oral and written.
- GS5.** Effective use read and interpret the data
- GS6.** read instructions, guidelines/procedures and reports
- GS7.** analyze the parameters and output
- GS8.** communicate effectively with team members.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health, Safety & Environment</i>	28	14	-	-
PC1. Perform Health and safety measures for installing and operating Green hydrogen system	2	1	-	-
PC2. Identify the requirements for safe work area at hydrogen generation project site	2	1	-	-
PC3. Understand health & Safety standards, procedures and regulations applicable for safe operation of hydrogen systems	2	1	-	-
PC4. Identify potential sources & causes of gas leaks, fire, explosion, bomb threatening, natural calamities etc.	2	1	-	-
PC5. Implement different detectors and safety tools	2	1	-	-
PC6. Review the Material Safety Data Sheet	2	1	-	-
PC7. maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.	2	1	-	-
PC8. Utilize sensors to alert the safety officer and other users when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals	2	1	-	-
PC9. Know how to administered first aid.	2	1	-	-
PC10. Identify the personal protective equipment used for the specific purpose.	2	1	-	-
PC11. Identify the hazards associated with hydrogen generation system	2	1	-	-
PC12. Carry out mock testing of firefighting system	2	1	-	-
PC13. discuss all applicable statutory requirements along with safety regulations in terms of fire protection.	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Incorporate good housekeeping practices and infection control guidelines.	2	2	-	-
<i>Commissioning Checklist</i>	22	11	-	-
PC15. Appreciate the importance of Commissioning activities	2	1	-	-
PC16. show how required activities according to commissioning checklist are performed	2	1	-	-
PC17. Know dismantling and removal of components	2	1	-	-
PC18. use the checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendations	2	1	-	-
PC19. explain Green hydrogen system installation, testing and commissioning checklist	2	1	-	-
PC20. Identify various Inputs, outputs and key performance metrics for hydrogen generation	2	-	-	-
PC21. Implement all pre-commissioning tests and other aspects including hydrogen purity, pressure test, leakage	2	1	-	-
PC22. Make compressor Assembly operational	2	1	-	-
PC23. Perform & test Drying Plant Hydraulic	1	1	-	-
PC24. Carry out safety tests	1	1	-	-
PC25. Perform a trial run before the commercial runs	2	1	-	-
PC26. commission Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards/SOPs etc	2	1	-	-
NOS Total	50	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4017
NOS Name	HSE and Commissioning Checklist
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

Qualification Pack

HYC/N4018: Roles & Responsibilities

Description

This unit covers the detailed roles & responsibilities of a green hydrogen plant technician.

Scope

The scope covers the following :

- Roles & responsibilities

Elements and Performance Criteria

• Roles & responsibilities

To be competent, the user/individual on the job must be able to:

- PC1.** Regularly inspect, troubleshoot, and maintain all machinery and equipment involved in hydrogen production, storage, and distribution.
- PC2.** Perform preventive maintenance tasks to minimize downtime and ensure equipment reliability, implementing predictive maintenance strategies using data analysis and monitoring tools to address potential issues proactively.
- PC3.** Conduct routine maintenance on electrolyzers for peak efficiency, monitoring and troubleshooting performance, including membrane degradation, gas leaks, and electrode fouling.
- PC4.** Coordinate with manufacturers or suppliers for specialized maintenance or component replacements.
- PC5.** Execute maintenance tasks on chemical equipment, pumps, valves, air compressors, water treatment RO's, fans, vessels, and buildings, including routine inspections, lubrication, and maintenance of hydrogen compressors, valves, seals, and pipeline.
- PC6.** Perform regular inspections of hydrogen storage equipment for integrity, safety valves, and instrumentation.
- PC7.** Conduct maintenance on cryogenic components, heat exchangers, and equipment in liquefaction units.
- PC8.** Participate in troubleshooting efforts to identify and resolve equipment malfunctions, working closely with other maintenance personnel to diagnose and address complex issues.
- PC9.** Stay updated on maintenance best practices and technologies, participating in training programs to enhance skills and knowledge.
- PC10.** Participate in continuous improvement initiatives to enhance the reliability and efficiency of maintenance processes, providing feedback on procedures and suggesting improvements.
- PC11.** Coordinate with other maintenance personnel to plan and execute scheduled maintenance activities, while also offering input on equipment performance and reliability for maintenance planning.
- PC12.** Adhere to and enforce strict safety protocols to ensure a safe working environment for all personnel, including the proper use of PPE.
- PC13.** Conduct safety inspections and participate in training programs, responding promptly to emergencies and incidents, taking appropriate actions to mitigate risks.

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- PC14.** Perform routine inspections and maintenance on the emergency diesel generator.
- PC15.** Operate facility mobile equipment and perform routine inspections and maintenance.
- PC16.** Support safety and environmental standards, inspections, and testing.
- PC17.** Participate in emergency response drills and be prepared to respond to incidents involving the release or handling of hydrogen, implementing emergency shutdown procedures when necessary.
- PC18.** Monitor and respond to gas detection alarms, ensuring proper ventilation in case of gas releases.
- PC19.** Calibrate and test mechanical equipment to maintain accurate and efficient operations.
- PC20.** Monitor and analyze data from various sensors to identify deviations and address potential problems.
- PC21.** Diagnose and troubleshoot technical issues related to electrolyzers and other equipment to minimize downtime and production interruptions.
- PC22.** Collaborate with other departments to identify root causes of problems and implement effective solutions. Conduct root cause analysis for equipment failures or process deviations, identifying and addressing underlying issues to prevent future occurrences.
- PC23.** Maintain accurate records of maintenance activities, including inspections, repairs, and equipment modifications.
- PC24.** Update and review documentation related to equipment manuals, procedures, and safety guidelines.
- PC25.** Work closely with engineers, operations teams, and other stakeholders to optimize plant performance, collaborating on projects aimed at improving efficiency, reducing costs, and enhancing safety.
- PC26.** Identify opportunities for process improvement and efficiency enhancements.
- PC27.** Implement and optimize predictive maintenance programs to maximize equipment reliability and lifespan.
- PC28.** Maintain accurate logs and records of operational parameters, events, and maintenance activities
- PC29.** Generating reports on equipment performance, incidents, and production metrics.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** procedures and instructions specific to the area of assignment
- KU2.** industrial maintenance practices, preferably in hydrogen production or a similar process industry
- KU3.** troubleshooting of machinery and equipment associated with hydrogen production, storage, and distribution.
- KU4.** implementation and optimization of predictive maintenance programs using advanced data analysis tools
- KU5.** cryogenic component maintenance, including routine inspections and preventive measures
- KU6.** safety protocols, emergency response, and hydrogen handling procedures.
- KU7.** equipment alignment and calibration

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- KU8.** collaboration with cross-functional teams
- KU9.** importance of effective communication with stakeholders
- KU10.** record-keeping, report generation, and meticulous documentation updates.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use Hand and Power Tools
- GS2.** communicate effectively and share the information efficiently
- GS3.** Effective use of computers (MS office tools)
- GS4.** Presentation skills -oral and written.
- GS5.** Effective use read and interpret the data
- GS6.** read instructions, guidelines/procedures and reports
- GS7.** analyze the parameters and output
- GS8.** communicate effectively with team members.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
• <i>Roles & responsibilities</i>	60	30	-	-
PC1. Regularly inspect, troubleshoot, and maintain all machinery and equipment involved in hydrogen production, storage, and distribution.	2	1	-	-
PC2. Perform preventive maintenance tasks to minimize downtime and ensure equipment reliability, implementing predictive maintenance strategies using data analysis and monitoring tools to address potential issues proactively.	2	1	-	-
PC3. Conduct routine maintenance on electrolyzers for peak efficiency, monitoring and troubleshooting performance, including membrane degradation, gas leaks, and electrode fouling.	2	1	-	-
PC4. Coordinate with manufacturers or suppliers for specialized maintenance or component replacements.	2	1	-	-
PC5. Execute maintenance tasks on chemical equipment, pumps, valves, air compressors, water treatment RO's, fans, vessels, and buildings, including routine inspections, lubrication, and maintenance of hydrogen compressors, valves, seals, and pipeline.	2	1	-	-
PC6. Perform regular inspections of hydrogen storage equipment for integrity, safety valves, and instrumentation.	3	1	-	-
PC7. Conduct maintenance on cryogenic components, heat exchangers, and equipment in liquefaction units.	2	1	-	-
PC8. Participate in troubleshooting efforts to identify and resolve equipment malfunctions, working closely with other maintenance personnel to diagnose and address complex issues.	2	1	-	-
PC9. Stay updated on maintenance best practices and technologies, participating in training programs to enhance skills and knowledge.	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Participate in continuous improvement initiatives to enhance the reliability and efficiency of maintenance processes, providing feedback on procedures and suggesting improvements.	2	1	-	-
PC11. Coordinate with other maintenance personnel to plan and execute scheduled maintenance activities, while also offering input on equipment performance and reliability for maintenance planning.	2	1	-	-
PC12. Adhere to and enforce strict safety protocols to ensure a safe working environment for all personnel, including the proper use of PPE.	2	1	-	-
PC13. Conduct safety inspections and participate in training programs, responding promptly to emergencies and incidents, taking appropriate actions to mitigate risks.	2	1	-	-
PC14. Perform routine inspections and maintenance on the emergency diesel generator.	2	1	-	-
PC15. Operate facility mobile equipment and perform routine inspections and maintenance.	3	2	-	-
PC16. Support safety and environmental standards, inspections, and testing.	2	1	-	-
PC17. Participate in emergency response drills and be prepared to respond to incidents involving the release or handling of hydrogen, implementing emergency shutdown procedures when necessary.	2	1	-	-
PC18. Monitor and respond to gas detection alarms, ensuring proper ventilation in case of gas releases.	2	1	-	-
PC19. Calibrate and test mechanical equipment to maintain accurate and efficient operations.	2	1	-	-
PC20. Monitor and analyze data from various sensors to identify deviations and address potential problems.	2	1	-	-
PC21. Diagnose and troubleshoot technical issues related to electrolyzers and other equipment to minimize downtime and production interruptions.	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Collaborate with other departments to identify root causes of problems and implement effective solutions. Conduct root cause analysis for equipment failures or process deviations, identifying and addressing underlying issues to prevent future occurrences.	2	1	-	-
PC23. Maintain accurate records of maintenance activities, including inspections, repairs, and equipment modifications.	2	1	-	-
PC24. Update and review documentation related to equipment manuals, procedures, and safety guidelines.	2	1	-	-
PC25. Work closely with engineers, operations teams, and other stakeholders to optimize plant performance, collaborating on projects aimed at improving efficiency, reducing costs, and enhancing safety.	2	1	-	-
PC26. Identify opportunities for process improvement and efficiency enhancements.	2	1	-	-
PC27. Implement and optimize predictive maintenance programs to maximize equipment reliability and lifespan.	2	1	-	-
PC28. Maintain accurate logs and records of operational parameters, events, and maintenance activities	2	1	-	-
PC29. Generating reports on equipment performance, incidents, and production metrics.	2	1	-	-
NOS Total	60	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4018
NOS Name	Roles & Responsibilities
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4.5
Credits	1.5
Version	6.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4001.Hydrogen Safety, Security and Health Management Procedures	35	60	-	-	95	10
HYC/N4014.Introduction to Green hydrogen and value chain	95	45	0	0	140	20
HYC/N4015.Water Electrolyser systems	90	45	0	0	135	20
HYC/N4016.Green Hydrogen Storage systems	40	20	0	0	60	20
HYC/N4017.HSE and Commissioning Checklist	50	25	0	0	75	10
HYC/N4018.Roles & Responsibilities	60	30	0	0	90	10
HYC/N9301.Working effectively in a team	20	30	-	-	50	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	410	285	0	0	695	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure

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NOS	National Occupational Standard(s)
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NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack

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KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.